Assignment 6.1

Week 6

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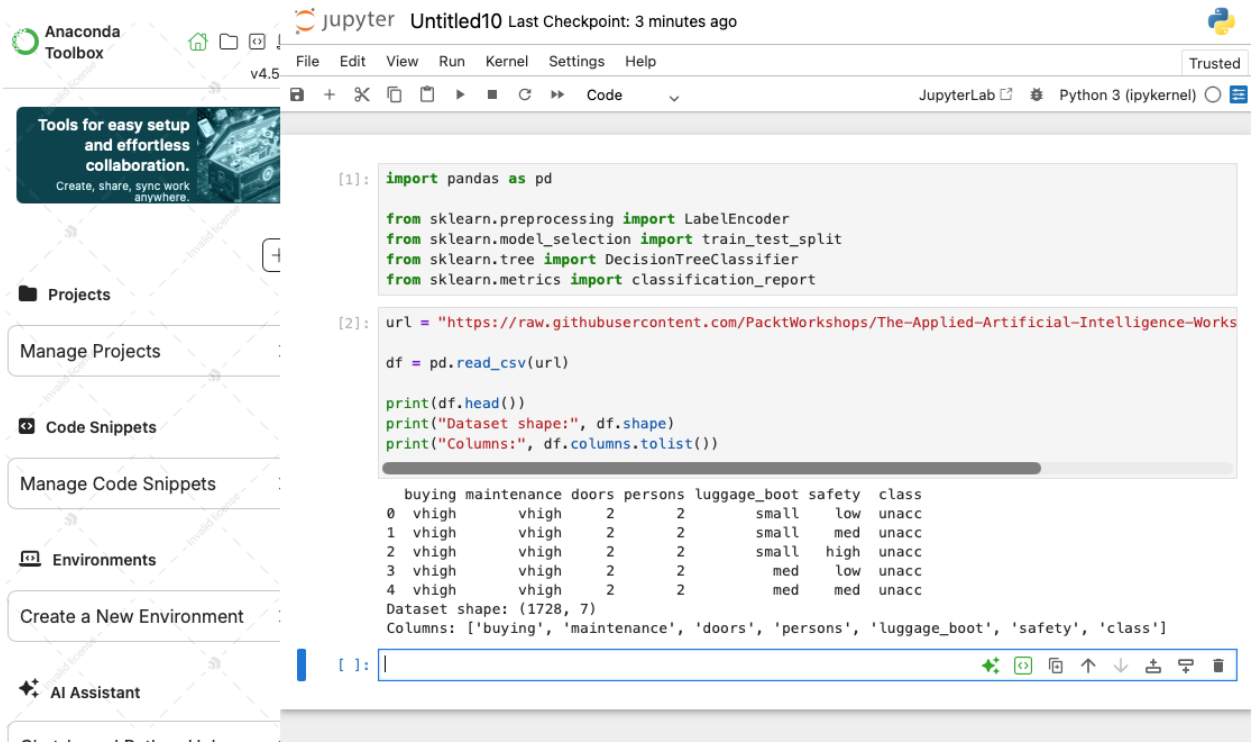
Professor Dr. Fuat Akal

Sep 11, 2026

Assignment 6.1: Decision Trees, Random Forests

Activity 4.01: Car Data Classification

Step 1 and 2



Codes:

```
import pandas as pd

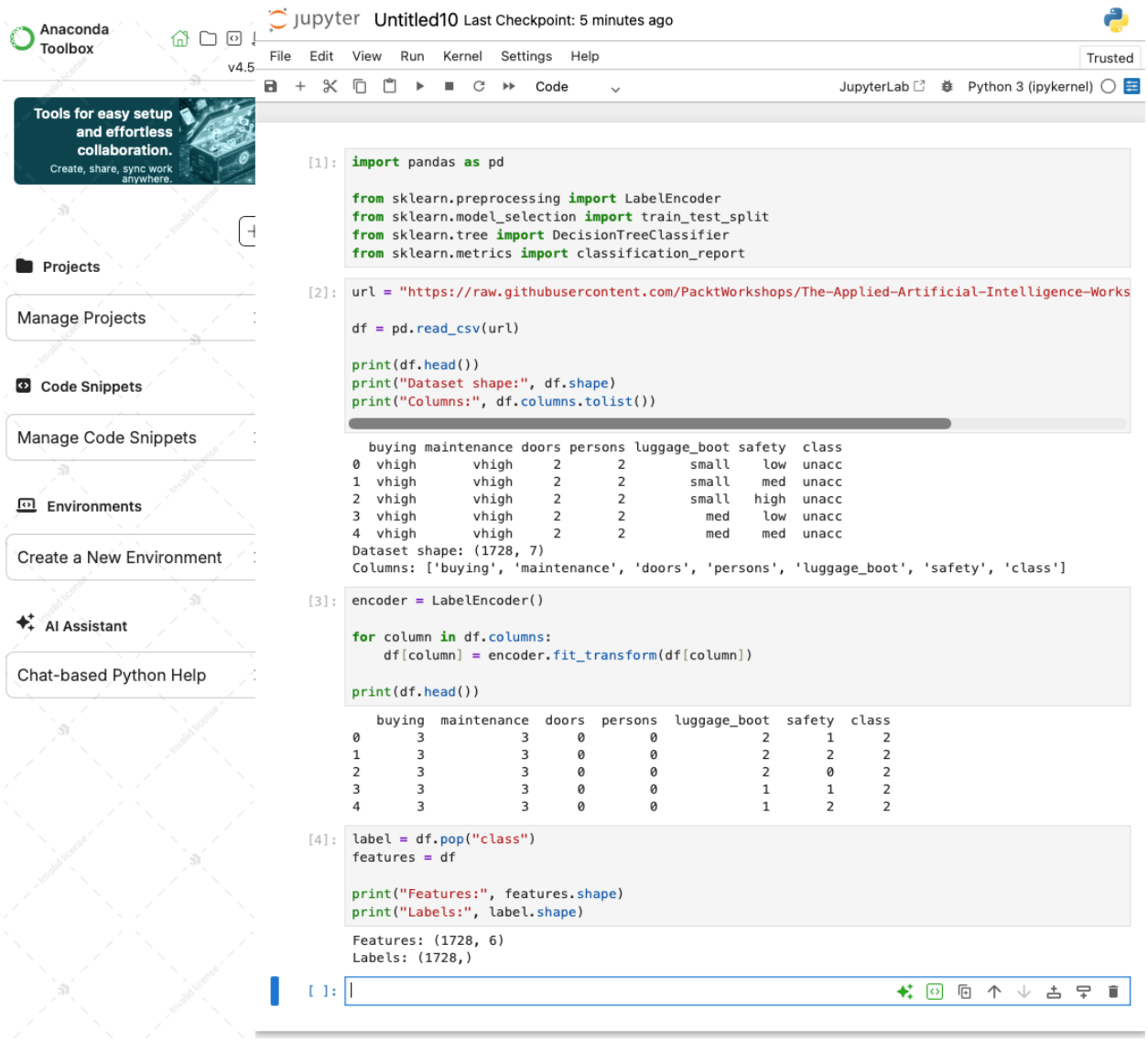
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report

url = "https://raw.githubusercontent.com/PacktWorkshops/The-Applied-Artificial-Intelligence-Workshop/master/Datasets/car.csv"

df = pd.read_csv(url)

print(df.head())
print("Dataset shape:", df.shape)
print("Columns:", df.columns.tolist())
```

Step 3 and 4



Codes:

```
encoder = LabelEncoder()

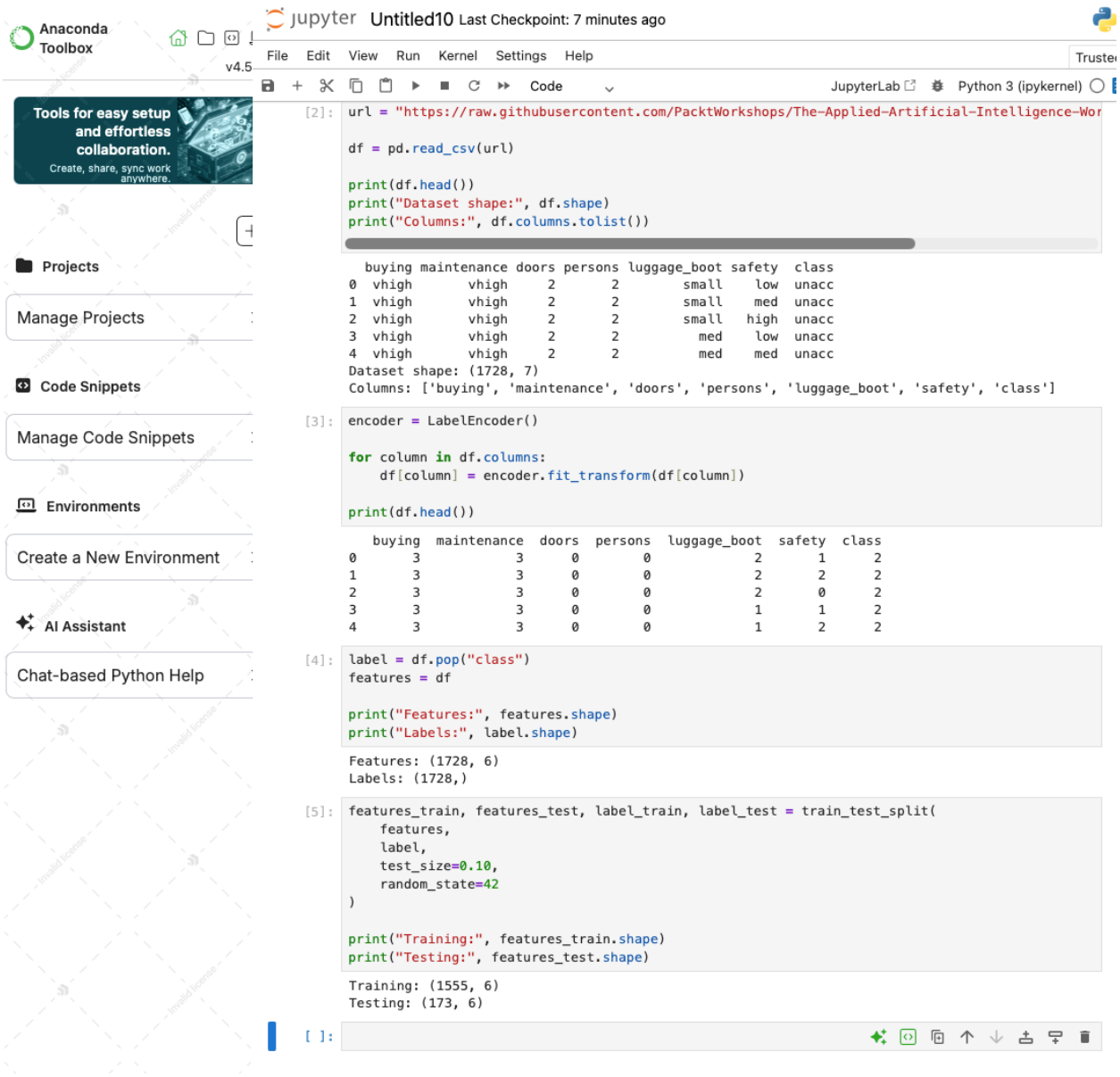
for column in df.columns:
   df[column] = encoder.fit_transform(df[column])

print(df.head())


label = df.pop("class")
features = df

print("Features:", features.shape)
print("Labels:", label.shape)
```

Step 5 (90% training data and 10% testing data).

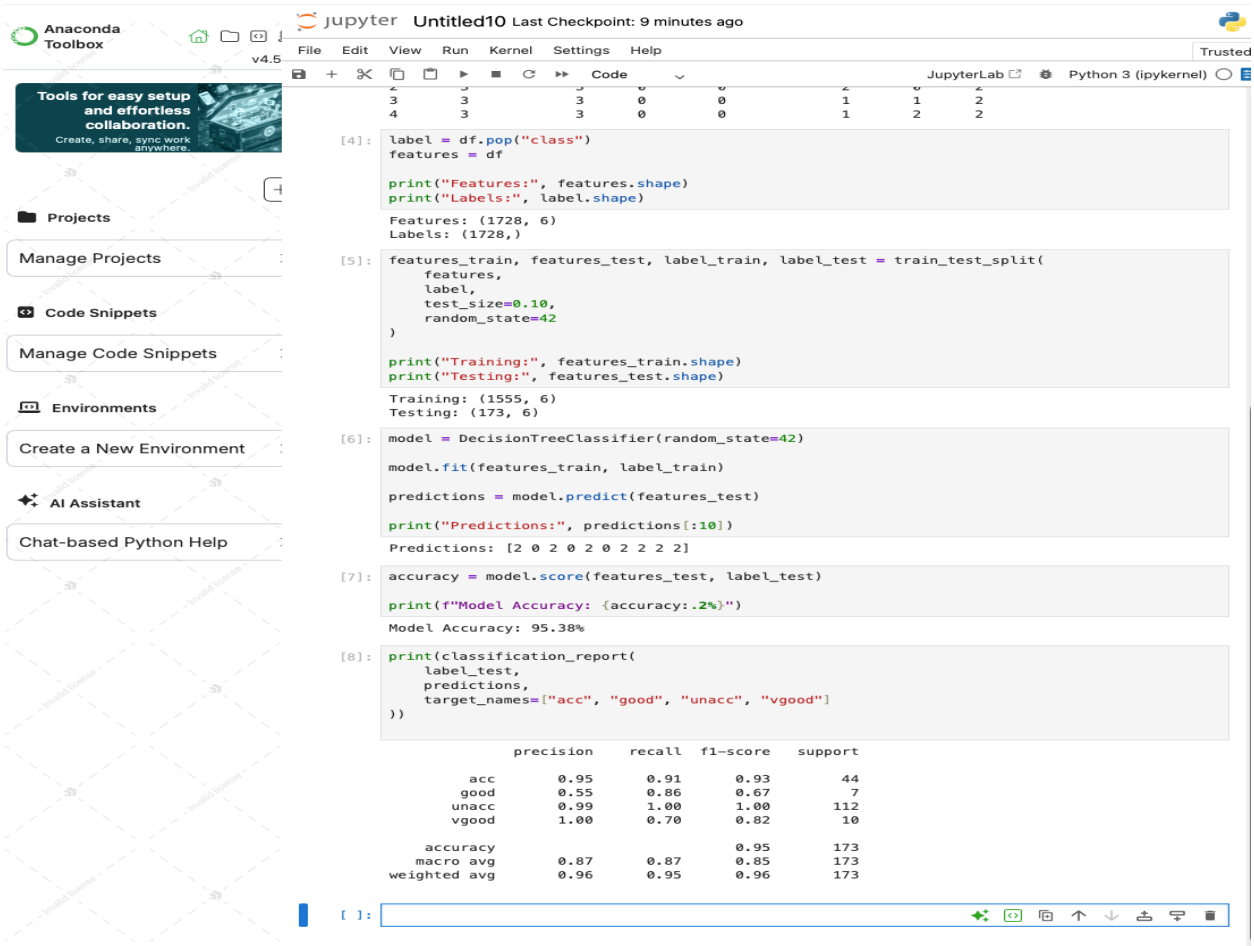


Code:

```
features_train, features_test, label_train, label_test = train_test_split(
   features,
   label,
   test_size=0.10,
   random_state=42
)

print("Training:", features_train.shape)
print("Testing:", features_test.shape)
```

Step 6, 7 and 8:



Codes:

```
model = DecisionTreeClassifier(random_state=42)

model.fit(features_train, label_train)

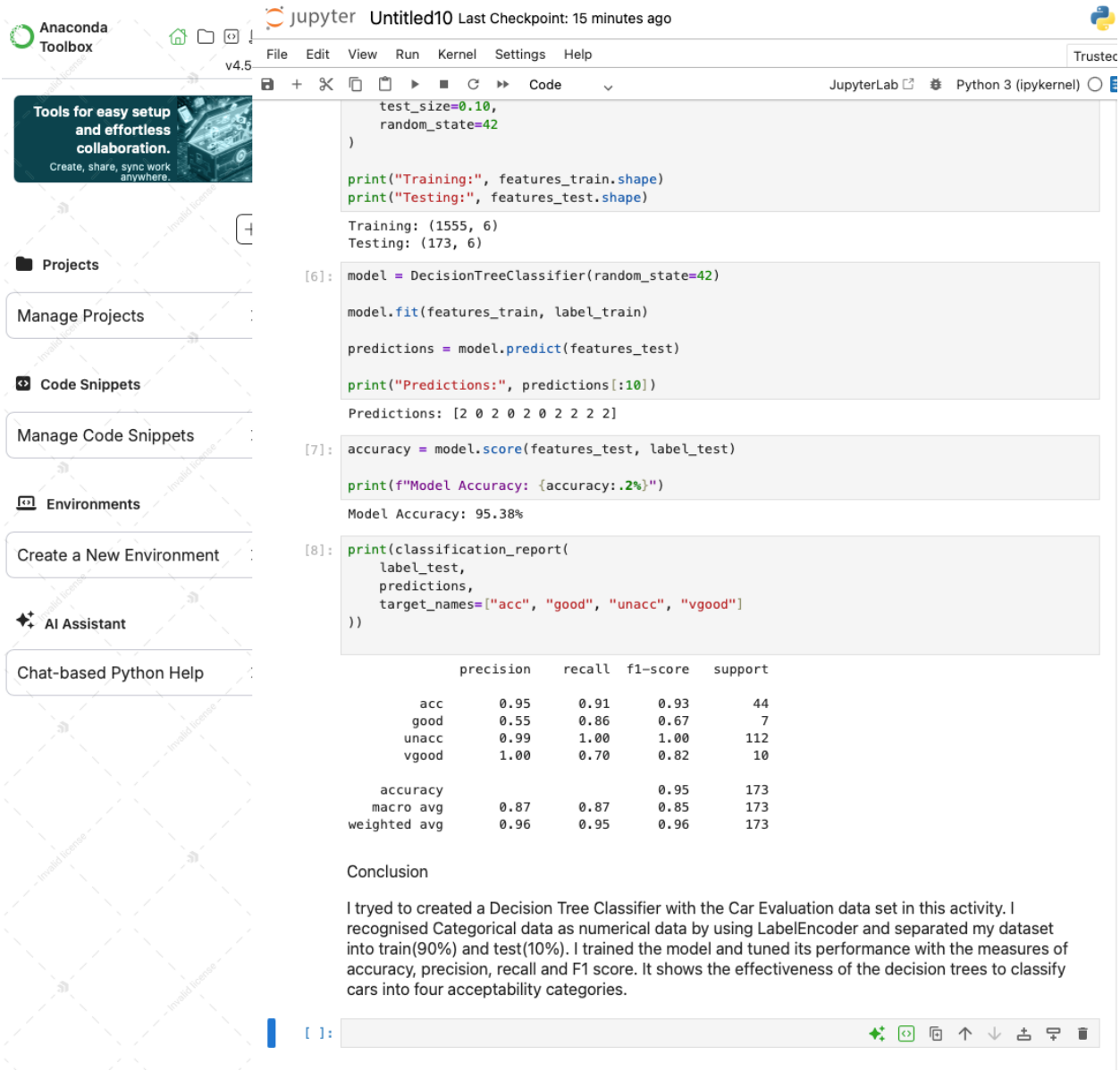
predictions = model.predict(features_test)

print("Predictions:", predictions[:10])


accuracy = model.score(features_test, label_test)

print(f"Model Accuracy: {accuracy:.2%}")

print(classification_report(
   label_test,
   predictions,
   target_names=["acc", "good", "unacc", "vgood"]
))
```


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Manage Code Snippets
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AI Assistant
Chat-based Python Help
jupyter Untitled10 Last Checkpoint: 15 minutes ago
File Edit View Run Kernel Settings Help
Code
JupyterLab
Python 3 (ipykernel)
test_size=0.10,
random_state=42
)
print("Training:", features_train.shape)
print("Testing:", features_test.shape)
Training: (1555, 6)
Testing: (173, 6)
[6]:
model = DecisionTreeClassifier(random_state=42)
model.fit(features_train, label_train)
predictions = model.predict(features_test)
print("Predictions:", predictions[:10])
Predictions: [2 0 2 0 2 0 2 2 2 2]
[7]:
accuracy = model.score(features_test, label_test)
print(f"Model Accuracy: {accuracy:.2%}")
Model Accuracy: 95.38%
[8]:
print(classification_report(
label_test,
predictions,
target_names=["acc", "good", "unacc", "vgood"]
))
precision recall f1-score support
acc 0.95 0.91 0.93 44
good 0.55 0.86 0.67 7
unacc 0.99 1.00 1.00 112
vgood 1.00 0.70 0.82 10
accuracy 0.95 173
macro avg 0.87 0.87 0.85 173
weighted avg 0.96 0.95 0.96 173
Conclusion
I tryed to created a Decision Tree Classifier with the Car Evaluation data set in this activity. I recognised Categorical data as numerical data by using LabelEncoder and separated my dataset into train(90%) and test(10%). I trained the model and tuned its performance with the measures of accuracy, precision, recall and F1 score. It shows the effectiveness of the decision trees to classify cars into four acceptability categories.
[ ]: